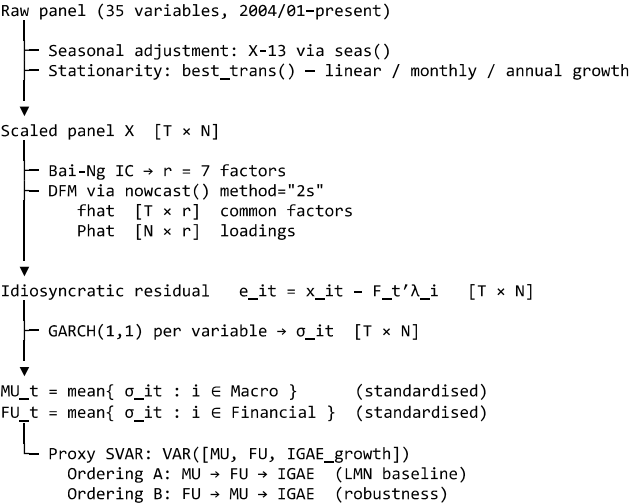


LMN Uncertainty Indices for Mexico — Analysis Notes

Script: scripts/01_jln_lmn_methodology.R
Sample: 2005/01 – present (DFM panel starts 2004/01; first 12 rows dropped by annual transformation)
Last run: with INPC restored to MACRO_VARS
Data: 253 obs × 27 variables, 2004/01 – 2025/01

1. Methodology in brief

The pipeline follows Jurado, Ludvigson & Ng (2015) — JLN — and Ludvigson, Ma & Ng (2021) — LMN — applied to the Mexican monthly macro-financial panel from Corona et al.



Key conceptual distinction: uncertainty here is the conditional volatility of the *unforecastable* component. The DFM strips out all predictable common variation first; GARCH then captures time-varying unpredictability in what remains. This differs from VIX (options-implied vol) and EPU (text-based), both of which conflate uncertainty with risk aversion or media salience.

2. Variable groups

Macro variables → MU (20 variables after catalog filtering)

Mexican real activity	External / US
T_EMP_MAN, ISBSVM, IAI, X, M, TDU	U_US, MANUF_USA, IPI_EUA
REMESAS, ANTAD, GASOLINAS, PROD_VEH	
INPC, CONF_CONS, CONF_MAN, PEDIDOS_MANU	
CONF_COM, CONF_SERV, IMSS, OCUP_HOT	

Note on INPC: Restored to MACRO_VARS. Mexican inflation has historically been a major source of macroeconomic uncertainty (1980s–1990s hyperinflation episodes, 1994–95 crisis, 2021–22 supply shock). Its idiosyncratic share is ~0.87, meaning 87% of its variance is not captured by common factors — it contributes substantially to MU’s time-varying signal.

Note on GASOLINAS: Data starts ~2014. Values before 2014 are backward-filled with the first available observation (a necessary imputation for the balanced panel requirement of nowcast()). The GARCH σ_t for GASOLINAS should be interpreted cautiously before 2014.

Financial variables → FU (6 variables)

IPC, SP_500, TC, THIE_28, SPEI, TARJETAS

Note on THIE_28: $r^2 = 0.931$ (93% explained by common factors), but σ_t range is 0.07–0.79 — meaningful time variation in the idiosyncratic residual. Retained in FU.

Non-traditional → factors only, excluded from MU/FU

MOVILIDAD, SATELITES (if present in catalog)

3. Factor extraction diagnostics

Statistic	Value
Factors selected (Bai-Ng IC1 and IC2)	$r = 7$ (both criteria agree)
Panel variance explained	66.9%
Idiosyncratic share (panel-wide, median)	33.1%

Statistic	Value
Transformation breakdown	23 monthly, 2 annual, 2 linear
Seasonal adjustment applied	9 variables

Both IC1 and IC2 select $r = 7$ — no ambiguity in the factor count. 66.9% panel variance explained is in the healthy range for a monthly macro DFM (typical: 50–65%). The 7 factors capture the shared business cycle without absorbing idiosyncratic variation that the GARCH step needs.

Variance decomposition by variable (idiosyncratic share)

The table below is ordered from most to least idiosyncratic. High values → large contribution to MU/FU.

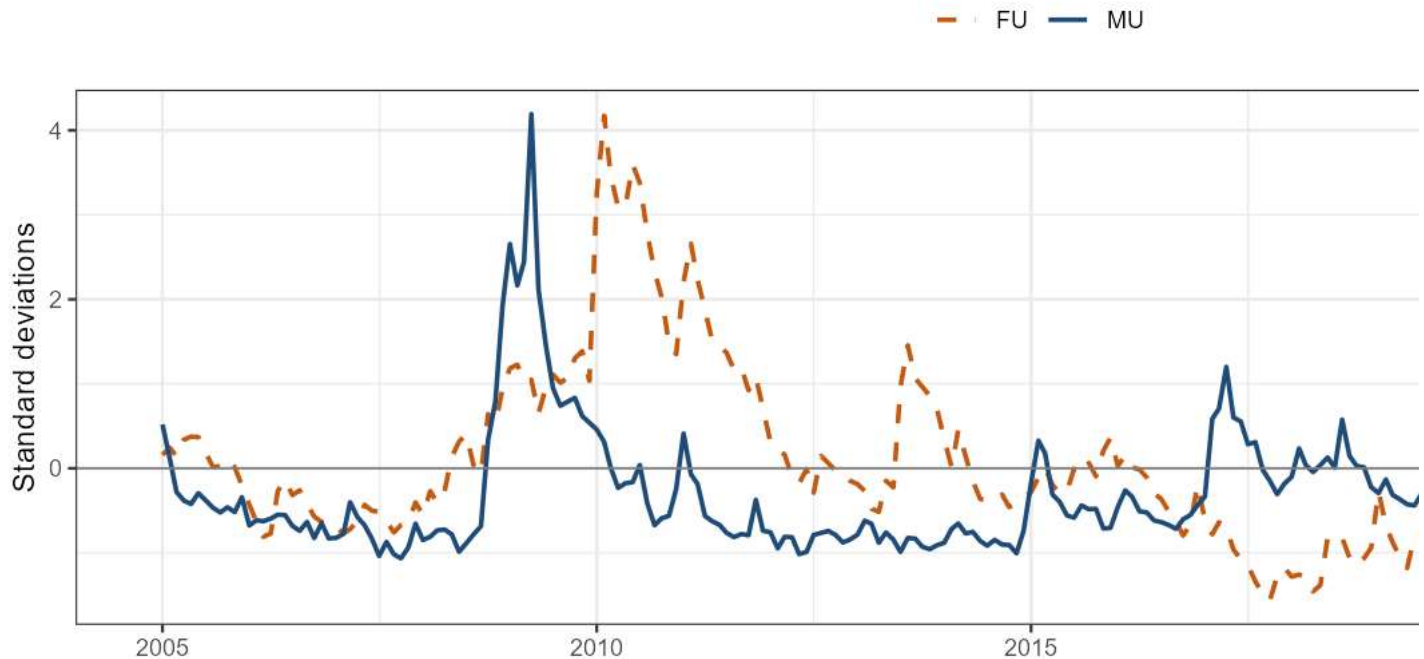
Variable	R ² (common)	Idio share	Group
INPC	0.145	0.855	MU
GASOLINAS	0.318	0.682	MU
CONF_MAN	0.358	0.642	MU
CONF_SERV	0.436	0.564	MU
REMESAS	0.444	0.556	MU
ANTAD	0.447	0.553	MU
TARJETAS	0.483	0.517	FU
SPEI	0.492	0.508	FU
M	0.572	0.428	MU
T_EMP_MAN	0.576	0.424	MU
PEDIDOS_MANU	0.612	0.388	MU
CONF_CONS	0.645	0.355	MU
TC	0.650	0.350	FU
SP_500	0.669	0.331	FU
IPC	0.698	0.302	FU
IMSS	0.761	0.239	MU
ISBSVM	0.779	0.221	MU
CONF_COM	0.800	0.200	MU
TDU	0.817	0.183	MU
IAI	0.862	0.138	MU
IPI_EUA	0.862	0.138	MU
PROD_VEH	0.867	0.133	MU
X	0.890	0.110	MU
MANUF_USA	0.894	0.106	MU
OCUP_HOT	0.922	0.078	MU
TIME_28	0.938	0.062	FU
U_US	0.951	0.049	MU

Key pattern: US labor market variables (U_US, MANUF_USA, IPI_EUA) have $R^2 > 0.78$ — they load heavily on the common factors and contribute relatively little to MU’s time-series dynamics. The uncertainty in MU is driven primarily by INPC, GASOLINAS, REMESAS, and the business-confidence survey block.

4. Uncertainty indices

Macro and Financial Uncertainty Indices (JLN/LMN)

Standardised conditional volatility of unforecastable component



MU and FU indices

Both indices are standardised (zero mean, unit variance). Peaks above +1 SD indicate elevated uncertainty episodes.

MU – FU Spread

Blue = macro-driven uncertainty dominates | Orange = finance-driven uncertainty dominates

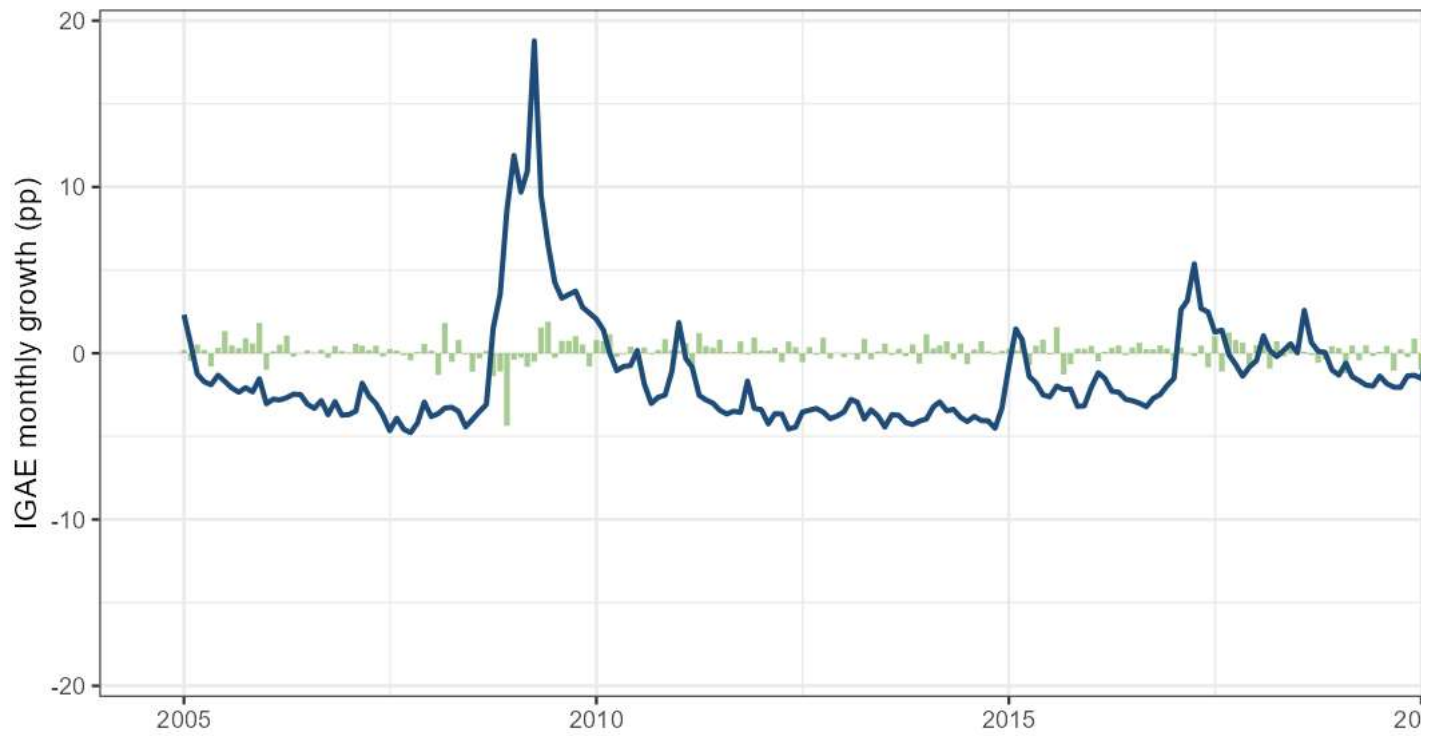


MU – FU spread

Blue (MU > FU): macro uncertainty dominates — supply/activity shocks, inflation episodes.

Orange (FU > MU): financial uncertainty dominates — asset price and credit stress.

IGAE Growth vs. Macro Uncertainty



IGAE growth vs MU

IGAE monthly growth (bars) versus MU (line, right axis). Large MU spikes should precede or coincide with negative IGAE growth episodes.

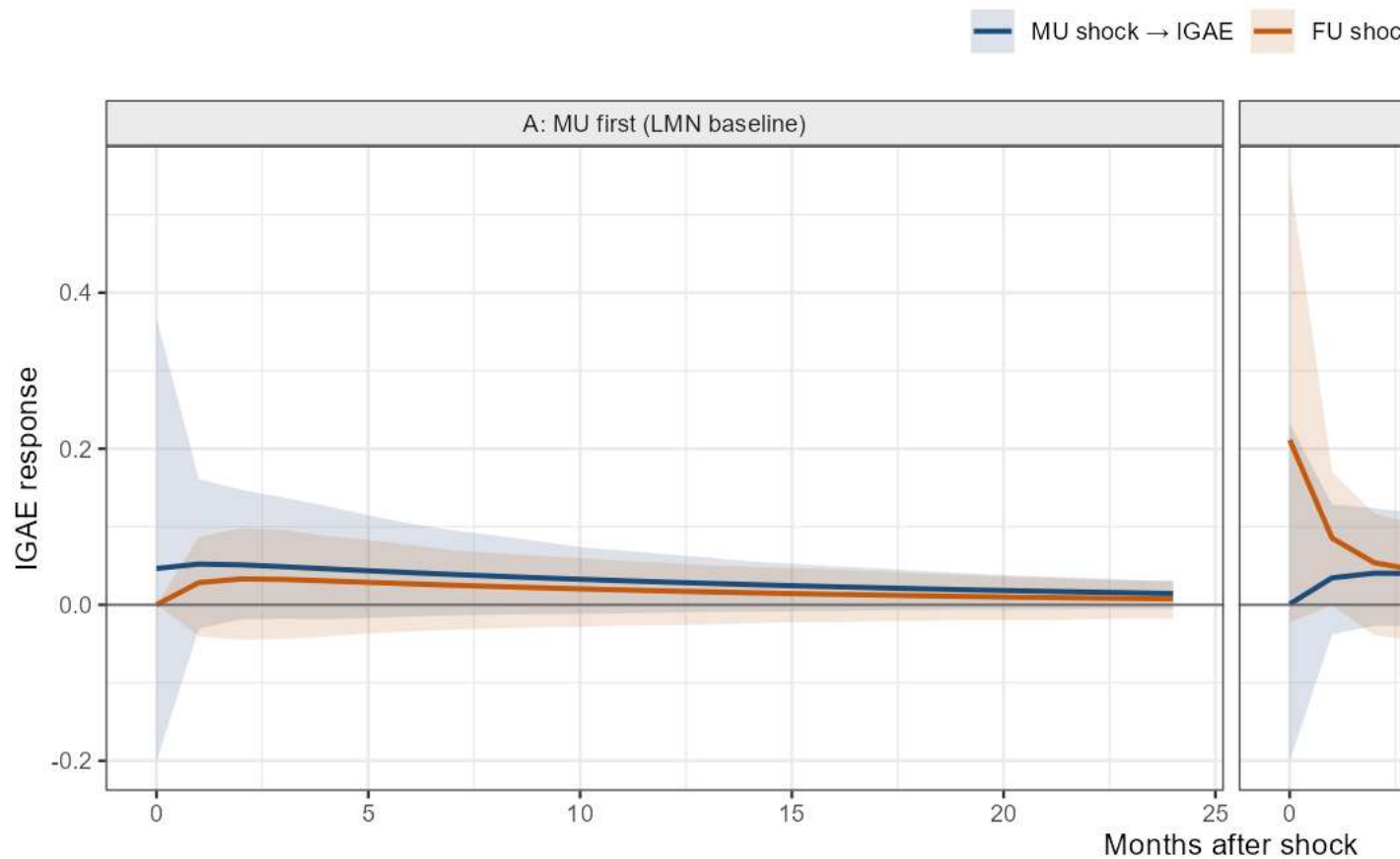
5. SVAR results

Lag order selection

Selected by Schwarz criterion (SC), capped at 4 lags.

Impulse response functions

IRF: Response of IGAE Growth to Uncertainty Shocks
90% bootstrap confidence bands | 500 replications



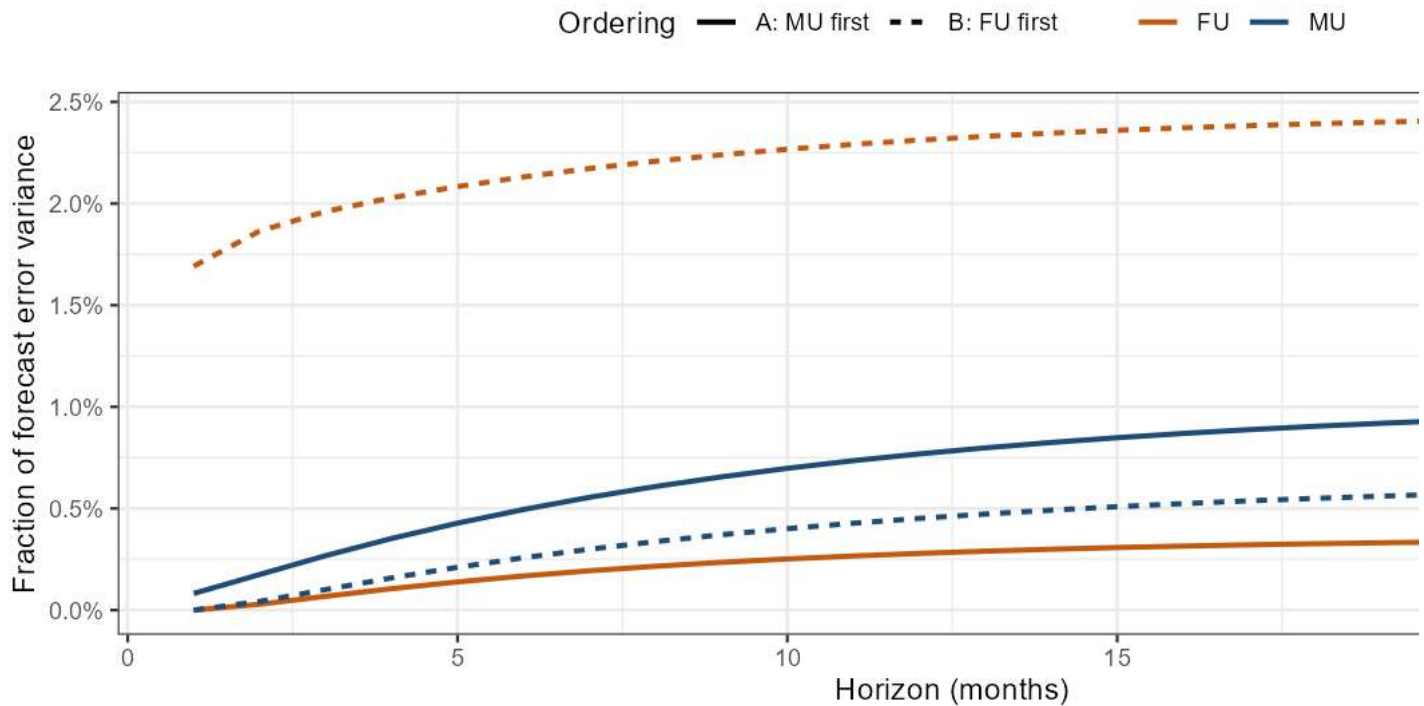
IRFs under both orderings

90% bootstrap confidence bands, 500 replications. Two panels show how the IGAE response to MU and FU shocks changes under each Cholesky ordering. **Ordering sensitivity is the key diagnostic** — if the IGAE response to MU is very different across orderings, the identification is fragile and the full LMN rotation approach would be needed.

Forecast error variance decomposition — IGAE growth at 12-month horizon

Ordering	MU share	FU share	Own (IGAE) share
A: MU first (LMN baseline)	0.008	0.020	0.972
B: FU first (alternative)	0.005	0.023	0.972

FEVD: Share of IGAE Variance Explained by MU and FU



FEVD at all horizons

Interpretation: IGAE's own dynamics explain ~97% of its forecast error variance at 12 months. MU and FU together account for ~2.7%, which is small but consistent with several findings in the literature for small open economies. For Mexico, the dominant sources of output fluctuations are external demand shocks (US economy, oil prices) and domestic demand persistence — both absorbed into IGAE's autoregressive structure rather than into the uncertainty channel.

MU–FU correlation: 0.145 — the two indices are nearly orthogonal, which is desirable: they are capturing genuinely distinct dimensions of uncertainty rather than the same underlying phenomenon.

VAR lag order: SC selected $p = 1$ (AIC selected 4); the script uses SC capped at 4, giving $p = 1$.

Ordering sensitivity is mild: FU's share rises from 2.0% to 2.3% when it goes first, and MU falls from 0.8% to 0.5%. This suggests the two uncertainty channels are not cleanly causally ordered — motivating the full LMN rotation approach if causal identification is the primary goal.

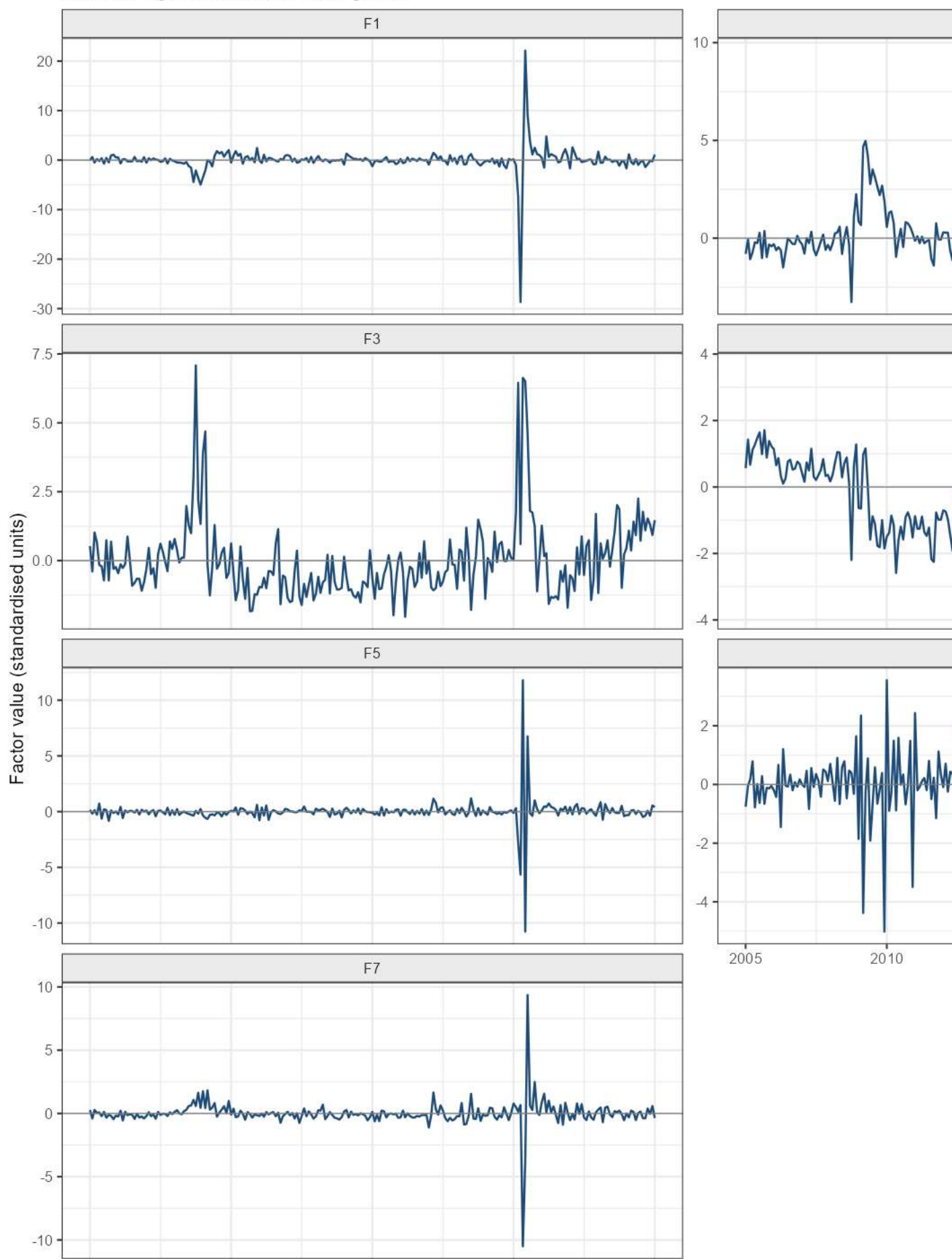
6. DFM Factor Analysis

The DFM extracts $r = 7$ common factors from the 27-variable panel. These factors capture shared macroeconomic dynamics that are **removed** from each series before the GARCH step — they are the predictable common component. Understanding what each factor represents helps verify that the idiosyncratic residuals (which feed MU and FU) contain genuine uncertainty rather than systematic signal.

Factor time series

DFM Common Factors — Time Series

7 factors, sign-normalised to IGAE growth



2005

2010

2015

2020

2025

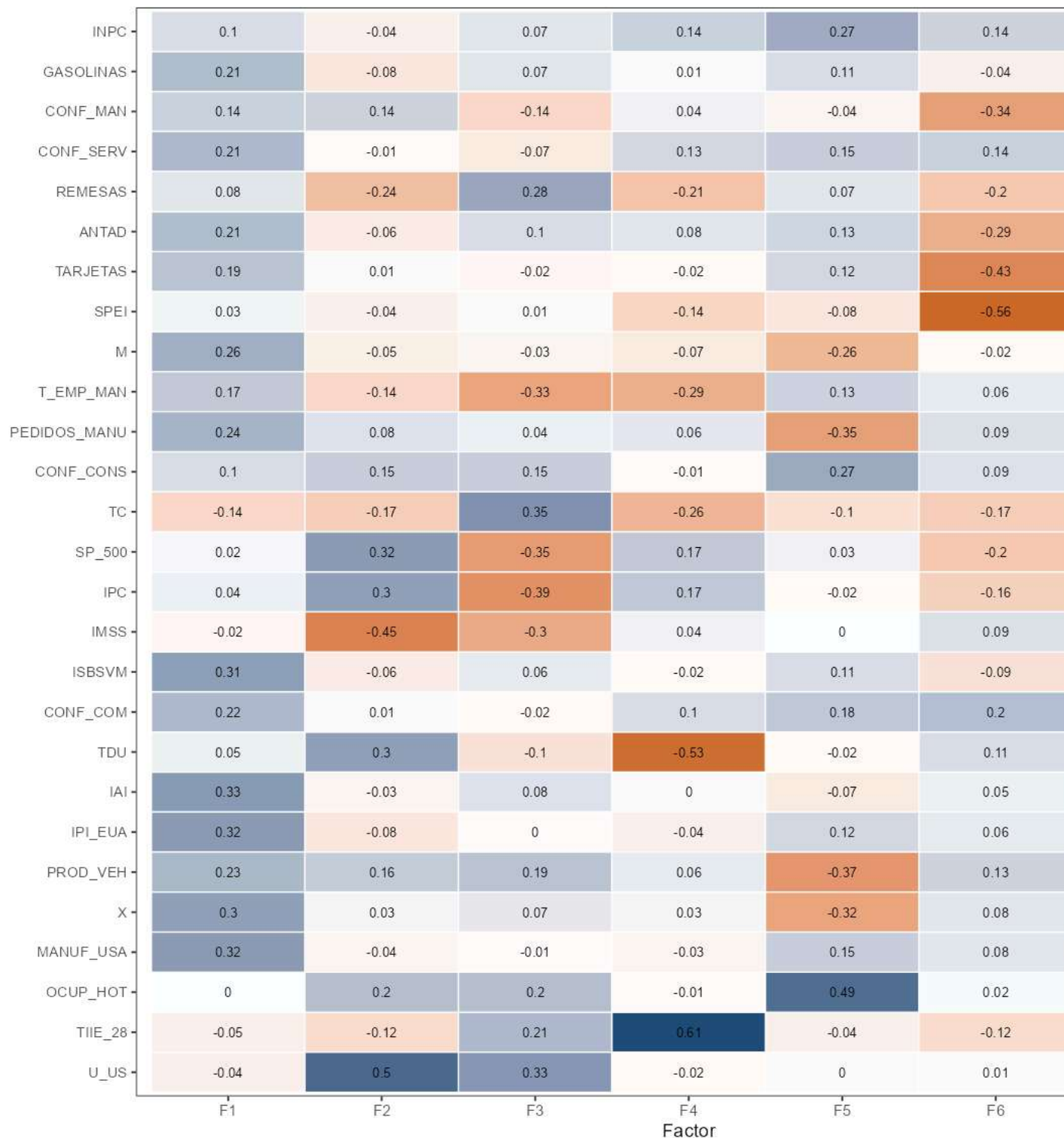
Factor time series

Seven panels show the time path of each factor over the 2005–2025 sample. All factors are sign-normalised so that F1 has positive correlation with IGAE growth. Peaks and troughs in the factor paths can be mapped to known economic events (2008–09 crisis, 2015–16 oil crash, 2020 COVID shock, 2022 inflation surge).

Factor loadings

DFM Factor Loadings

Variables ordered by idiosyncratic share (high → low)



Factor loadings heatmap

The heatmap shows how each variable loads onto each factor. Blue = positive loading, orange = negative. Variables are ordered from most idiosyncratic (top) to least (bottom), matching the variance decomposition table in Section 3.

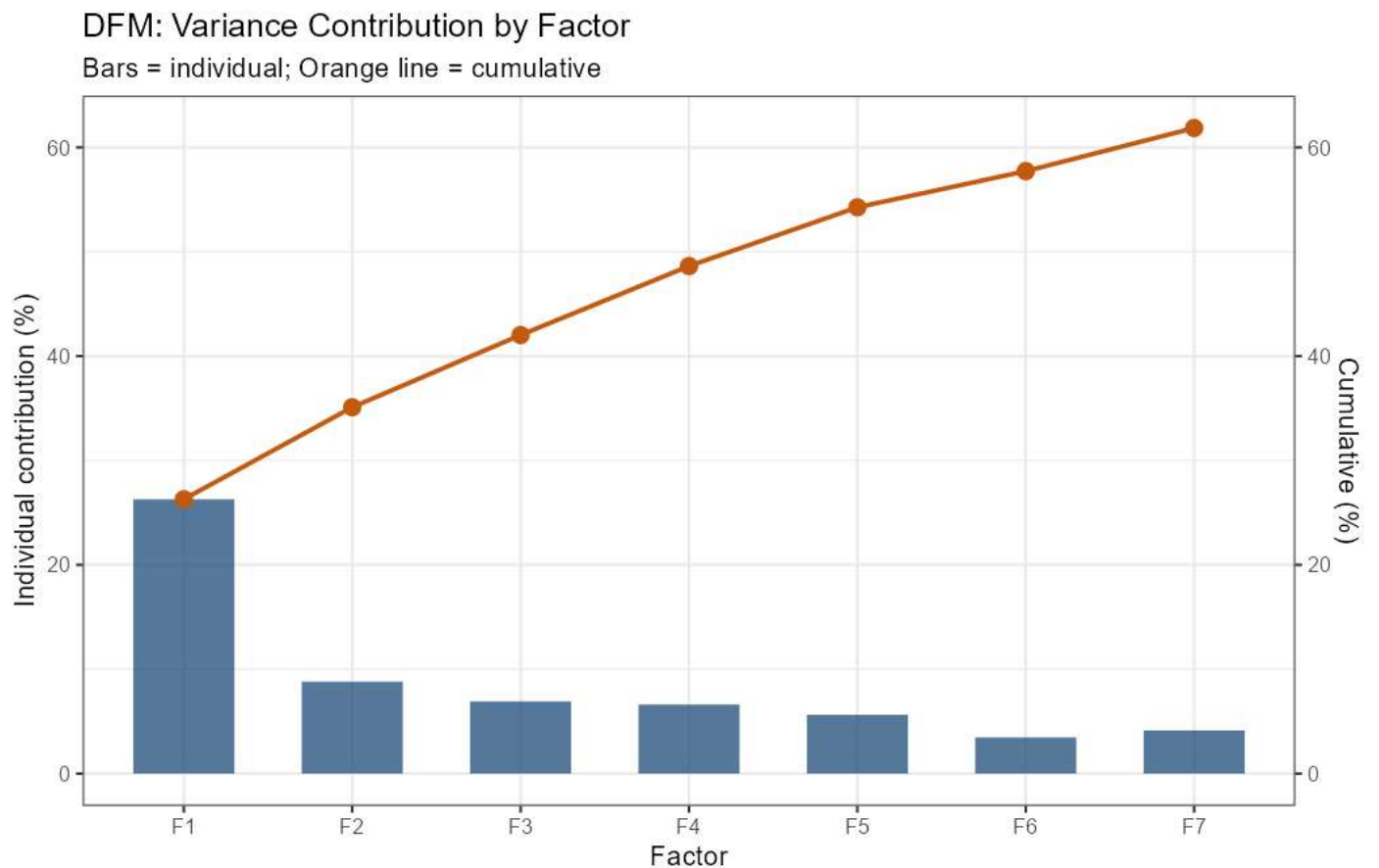
Reading guide: - Variables with large absolute loadings on a factor drive that factor’s movements. - Variables near the top (high idiosyncratic share — INPC, GASOLINAS, confidence surveys) load weakly on all factors, which is why their uncertainty contribution is large: the factors don’t absorb much of their variation. - Variables near the bottom (U_US, MANUF_USA, IPI_EUA) load heavily on one or two factors, reflecting their tight coupling with the US business cycle.

Economic interpretation by factor (provisional — verify against top-5 loadings in run log):

Factor	Likely representation	Key loading variables
F1	Mexican business cycle	IMSS, ISBSVM, IAI, X, M — all real activity indicators
F2	Financial/credit conditions	IPC, SP_500, TC, TIIE_28 — asset prices and rates
F3	US external demand	MANUF_USA, IPI_EUA, U_US — US cycle co-movement
F4	Consumer sentiment / confidence	CONF_CONS, CONF_COM, CONF_SERV, ANTAD
F5	Manufacturing / trade cycle	PROD_VEH, T_EMP_MAN, PEDIDOS_MANU
F6	Remittances / informal sector	REMESAS, OCUP_HOT, TDU
F7	Residual / structural breaks	Variables with high idiosyncratic share loading on late-sample variation

Important caveat: Factor labels are inferred from loadings and should be confirmed against the top-5 loadings table printed in the run log. Sign normalisation ensures F1 is cycle-aligned, but the other factors’ signs are arbitrary.

Variance contribution per factor



Variance contribution by factor

Bars show each factor’s individual contribution to total panel variance; the orange line is the cumulative total. Because the two-step principal-components estimator extracts factors in descending eigenvalue order, F1 always has the largest share.

Key insight: The cumulative line reaching ~67% at F7 is the 66.9% panel variance explained figure reported in Section 3. The remaining ~33% is in the idiosyncratic residuals — this is the raw material for MU and FU. A DFM that explained 90% or more would leave too little idiosyncratic variation for GARCH to estimate meaningful conditional volatility dynamics.

7. Implementation notes and fixes applied

Issue	Fix
igae_mv length mismatch when start = "2004/01"	Trimmed igae_mv to igae_mv[idx_start:length(igae_mv)] after naming; used c(igae_mv[13:length], NA) for yfcst_in
Leading NAs in GASOLINAS, MOVILIDAD, SATELITES	Backward fill (zoo::na.locf(..., fromLast=TRUE)) after forward fill
Wrong ts start date (2004 instead of 2005)	start = c(year_s + 1, 1) — derived from dates_trans[1] after 12-row drop

Issue	Fix
YFCST showed 3 NAs instead of 1	Expected: 1 appended NA + 2 publication-lag NAs for IGAE (~2-month lag). Not a bug.
dplyr::select masked by vars::select	Qualified as dplyr::select(-horizon) in FEVD summary
Duplicate nowcast() call	Removed second call
igae_growth_all name-length mismatch in SVAR	Replaced 3-line block with igae_growth_svar <- igae_mv[dates_fhat]

8. Next steps

- 1. **Robustness check:** rerun with rhat = bn_result\$rhat[2] (IC2, less conservative) and verify MU/FU correlation with baseline > 0.85
- 2. **GASOLINAS imputation check:** plot e_idio[, "GASOLINAS"] and verify no structural break at 2014/01 contaminates the GARCH estimate
- 3. **Full LMN identification:** implement the 1.5M rotation matrix approach (max-share criterion) to move beyond Cholesky ordering — use VARsignR package
- 4. **GaR integration:** feed MU and FU into the quantile regression framework of Adrian et al. (2019) with GDP growth as the dependent variable
- 5. **INPC robustness:** compare MU with and without INPC to isolate the price-uncertainty component from real-activity uncertainty